

Variance Retention as a Diagnostic Complement to Persistence-Relative Skill in Multi-Horizon PM₁₀ Forecasting

Francisco Federico Garcia Crespi

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Abstract

Persistence-relative RMSE skill is commonly used in multi-horizon point-forecast evaluation, but positive skill under this criterion does not ensure that a forecast preserves variability relevant to event detection. We evaluate multi-horizon PM₁₀ point forecasts under a rolling-origin protocol with persistence as an explicit baseline, comparing LightGBM and SARIMA across lead times $h \in \{1, 6, 24, 48\}$ hours at an hourly-resolution station in Madrid, Spain. The leakage-free rerun does not reproduce the previously reported high skill: LightGBM skill ranges from -0.079 to 0.073 , whereas SARIMA skill ranges from 0.031 to 0.132 . Variance retention nevertheless changes the interpretation of the positive SARIMA results: it falls from 79.8% at $h = 1$ to 2.9% and 0.4% at $h = 24$ and 48 , respectively. We report horizon-wise variance retention alongside RMSE-based skill and use Skill_{VP} , a bounded auxiliary diagnostic that discounts nominal skill by variance retention, to summarize this decoupling in a single horizon-wise quantity. Event-oriented evaluation under fold-recalibrated P75 thresholds confirms the horizon dependence: SARIMA recall falls from 0.785 at $h = 1$ to 0.099 at $h = 24$ and zero at $h = 48$. In this case study, small positive RMSE-relative skill does not by itself support claims of practical usefulness; amplitude and event-detection diagnostics materially qualify its interpretation.

Keywords: PM₁₀ forecasting; multi-horizon forecasting; rolling-origin evaluation; persistence baseline; forecast skill; variance retention; dynamic fidelity

1 Introduction

Short-term air-quality forecasting is used to anticipate changes in pollutant concentrations at successive lead times [Donnelly et al., 2015]. Its evaluation should therefore preserve the horizon as an explicit dimension: a forecast may improve on a simple reference at one lead time but not at another. Persistence is a transparent reference forecast and an established standard against which relative skill can be interpreted [Murphy, 1992].

A practical complication is that squared-error criteria (MSE/RMSE) dominate both model optimization and the reporting of multi-horizon performance. Under limited predictability, point forecasts that shrink toward a central tendency and dampen fluctuations can reduce squared error by avoiding large misses, even if this also suppresses variability that may be operationally relevant [Gneiting, 2011]. This trade-off is well known in point forecasting under squared loss: improvements in average error can coincide with reduced amplitude, and this tension may become more salient as lead time increases and the conditional distribution of outcomes broadens.

The gap addressed in this paper is empirical and protocol-specific. In multi-horizon PM₁₀ forecasting evaluated with rolling-origin protocols, positive persistence-relative RMSE skill can coexist with degraded dynamic fidelity, reflected by a pronounced reduction in predicted variance relative to observed variance at the same horizon. In such cases, a horizon-wise skill curve may remain positive at medium leads while the corresponding prediction sequences become under-variable and less responsive to episode evolution. Rolling-origin evaluation is central to this diagnostic because it aggregates performance over a sequence of forecast origins spanning changing conditions, thereby exposing horizon-dependent behaviors that may be less visible under a single blocked split [Tashman, 2000].

This study provides a bounded empirical characterization of this decoupling for LightGBM and SARIMA evaluated against a persistence baseline. We report horizon-wise RMSE and persistence-relative skill together

with observed and predicted variances and variance retention, and we introduce Skill_{VP} as a simple diagnostic auxiliary metric to summarize how nominal skill and variance retention jointly behave at each horizon. The pattern in which persistence-relative RMSE skill remains positive while predictive variance is substantially attenuated is sometimes described informally in the forecasting community as “ghost skill”; we use the term here only as shorthand for that specific decoupling and do not adopt it as formal terminology.

Contributions. The contribution is intentionally bounded and empirical:

- We provide horizon-wise evidence, under rolling-origin evaluation with persistence as an explicit baseline, that persistence-relative RMSE skill can decouple from dynamic fidelity as measured by variance retention. LightGBM and SARIMA serve as contrasting cases: LightGBM does not consistently beat persistence, while SARIMA’s modest positive skill at longer horizons coincides with severe variance attenuation and weak P75-event detection.
- We introduce Skill_{VP} as a simple diagnostic auxiliary metric that adjusts nominal persistence-relative RMSE skill by variance retention for horizon-wise interpretation, and we report event-oriented complements (recall, precision, flag rate) under fold-recalibrated thresholds as a further check on event sensitivity.

2 Related Work and Positioning

This manuscript is positioned as a restrained empirical-diagnostic study in multi-horizon environmental forecasting. It does not aim to advance general forecasting theory or introduce new universal verification metrics. Instead, it operationalizes several established verification ideas in a deterministic PM₁₀ case study using a rolling-origin protocol, persistence as an explicit baseline, horizon-by-horizon reading of skill, and an explicit diagnosis of variance retention.

2.1 Forecast skill and baseline-aware evaluation

The horizon-wise perspective motivates treating the horizon as the primary unit of interpretation in multi-step evaluation, rather than collapsing performance into a single aggregate number [Ben Taieb et al., 2012]. In applied environmental forecasting, this framing is paired with explicit reference forecasts that provide an operationally meaningful benchmark for interpreting whether a model adds information beyond simple persistence-like behavior [Murphy, 1992].

The present paper adopts this baseline-aware, horizon-wise reading in a deliberately narrow setting: deterministic point forecasts for PM₁₀ evaluated under a rolling-origin protocol with persistence as an explicit baseline. Rolling-origin aggregation preserves temporal ordering and exposes horizon-dependent behavior across repeated forecast origins [Tashman, 2000].

2.2 Variance-fidelity diagnostics

A well-known concern in deterministic point forecasting is that squared-error optimization can encourage shrinkage toward conditional means, which may suppress predictive variability as lead time increases [Gneiting, 2011]. In environmental and hydrological verification, this issue connects to antecedents that explicitly separate different components of performance, including diagnostics that track variability ratios (e.g., the variability component of KGE-style decompositions) and adjacent discussions of oversmoothing and amplitude damping [Gupta et al., 2009, Kling et al., 2012].

The ingredients used here are classical: a baseline-relative skill quantity and a variance ratio. Letting $\text{Skill}(h)$ denote persistence-relative RMSE skill at horizon h and letting $\sigma_y^2(h)$ and $\sigma_{\hat{y}}^2(h)$ denote the horizon-wise predicted and observed variances over the rolling-origin verification set, we define

$$\text{Skill}_{VP}(h) = \text{Skill}(h) \frac{\sigma_{\hat{y}}^2(h)}{\sigma_y^2(h)}, \quad (1)$$

used here in a bounded form (see Section 3.7). Skill_{VP} is positioned only as an auxiliary diagnostic complement that down-weights nominal persistence-relative RMSE skill when variance retention is low; its precise scope and boundaries are stated in Section 3.7.

2.3 Contribution statement

The strongest contribution of this paper is not conceptual novelty, but the empirical combination of a PM_{10} case study with rolling-origin evaluation, persistence as an explicit operational baseline, horizon-by-horizon performance reading, and an explicit variance-retention diagnosis. In the reported results, LightGBM and SARIMA serve as contrasting cases: LightGBM does not consistently improve on persistence, while SARIMA provides the clearer case in which modest positive persistence-relative RMSE skill at long horizons coexists with near-collapsed predictive variance. This contrast motivates interpreting skill curves jointly with variance retention, without treating either diagnostic as direct evidence of usefulness in a specific decision setting and without claiming that such decoupling must occur universally.

3 Methodology

3.1 Forecasting setting and multi-horizon notation

Let y_t denote the observed PM_{10} concentration at time index t . At each forecast origin t , a point forecasting model issues multi-horizon predictions $\hat{y}_{t+h|t}$ for horizons $h \in \{1, \dots, H\}$. Evaluation is performed separately for each horizon to avoid conflating short- and long-lead behavior, since both error characteristics and the operational interpretation of forecasts can change substantially with lead time.

3.2 Rolling-origin evaluation

The empirical analysis was conducted on an hourly PM_{10} series from the Casa de Campo monitoring station in Madrid, Spain, covering the full calendar year 2023 (2023-01-01 00:00 to 2023-12-31 23:00). The regular hourly grid contains 8,760 time stamps, of which 8,465 have measurements marked as valid by the data provider; invalid or absent targets are excluded from verification. The data were obtained from the Madrid Open Data portal operated by the Ayuntamiento de Madrid. The complete grid and the valid-observation count are reported separately because missing or provider-invalid values remain missing. Table 1 summarises the study site, data coverage, and evaluation protocol.

We adopt a rolling-origin (walk-forward) evaluation protocol to approximate sequential use. The design uses five expanding folds. The first training window contains 4,380 hourly grid points (50% of the year); each subsequent test window contains 876 grid points, and each completed test window is added to the next training window. Models are refitted at the start of each fold using only its training window. Within a test window, forecasts are issued sequentially using only information available at origin t ; the SARIMA state is updated without parameter refitting when a new observation becomes available. For each horizon h , verification aggregates over the set of valid origins $\mathcal{T}_h$ with targets y_{t+h} available, yielding horizon-wise sample sizes $N_h = |\mathcal{T}_h|$. No scaling is applied. Unavailable inputs are forward-filled only from the latest value available at the origin; backward filling is not used. The candidate and persistence forecasts are evaluated on exactly the same valid (t, h) verification pairs. This protocol avoids look-ahead and exposes the evaluation to changes in conditions across origins [Tashman, 2000].

3.3 Reference forecast: persistence

Persistence is used as an explicit baseline forecast. For each origin t and horizon h , the persistence prediction is

$$\hat{y}_{t+h|t}^{(p)} = y_t. \quad (2)$$

All skill statements in this paper are made relative to persistence and are reported horizon-wise.

Table 1: Summary of the study site, data coverage, and evaluation protocol.

Item	Value
Study site / station	Casa de Campo (Madrid, Spain)
Period	2023-01-01 00:00 to 2023-12-31 23:00
Temporal resolution	1 hour
Hourly grid / valid values	8,760 / 8,465
Data source	Madrid Open Data, magnitude 10, station 024
Forecast horizons (h)	1, 6, 24, 48 h
Rolling-origin folds	5 (expanding window, initial 50%)
Holdout split	80/20 chronological
Models compared	LightGBM; SARIMA
Baseline	Persistence

3.4 Empirical models: LightGBM and SARIMA

We report results for two empirical point-forecasting models. The first is a gradient-boosted decision tree model (LightGBM), trained independently for each fold and horizon h using current and lagged values at 0, 1, 6, 24, 48, and 168 hours, causal rolling means and standard deviations over 6, 24, 48, and 168 hours, and known target-calendar covariates [Ke et al., 2017]. The fixed configuration uses 200 trees, maximum depth 8, 31 leaves, learning rate 0.05, feature subsampling of 0.9, and random seed 42. The row-subsample parameter is 0.9 but its frequency remains at the LightGBM default of zero, so row subsampling is inactive. Fitting is deterministic and single-threaded. Unavailable inputs are forward-filled from the most recent valid observation; backward filling is never used, and invalid targets are excluded. LightGBM is included as a widely used nonlinear predictor in applied forecasting, not as a tuned state-of-the-art benchmark.

The second is SARIMA, a classical time-series model fitted on the raw PM_{10} series at the start of each rolling-origin fold using the pre-specified order $(p, d, q)(P, D, Q)_s = (1, 0, 1)(1, 0, 0)_{24}$, a constant term, and recursive multi-step forecasting [Box et al., 2015]. SARIMA is included not as a comprehensive statistical benchmark, but as a comparator that can exhibit qualitatively different dynamic-fidelity behavior under the same rolling-origin evaluation and persistence-relative skill framing.

3.5 Horizon-wise RMSE and persistence-relative RMSE skill

For model m at horizon h , the rolling-origin RMSE is

$$\text{RMSE}_m(h) = \sqrt{\frac{1}{N_h} \sum_{t \in \mathcal{T}_h} \left(\hat{y}_{t+h|t}^{(m)} - y_{t+h} \right)^2}. \quad (3)$$

Persistence-relative RMSE skill is defined as

$$\text{Skill}_{\text{RMSE}}(h) = 1 - \frac{\text{RMSE}_m(h)}{\text{RMSE}_p(h)}. \quad (4)$$

Positive values indicate lower RMSE than persistence at horizon h .

3.6 Dynamic fidelity via variance retention

To characterize amplitude preservation, we compute horizon-wise sample variances over the same evaluation set $\mathcal{T}_h$:

$$\text{Var}_{\text{obs}}(h) = \text{Var}(\{y_{t+h} : t \in \mathcal{T}_h\}), \quad \text{Var}_{\text{pred}}(h) = \text{Var}\left(\{\hat{y}_{t+h|t}^{(m)} : t \in \mathcal{T}_h\}\right). \quad (5)$$

Variance retention is reported as a percentage,

$$\text{VR}(h) = 100 \times \frac{\text{Var}_{\text{pred}}(h)}{\text{Var}_{\text{obs}}(h)}. \quad (6)$$

Values substantially below 100% indicate that predictions are under-variable relative to observations at the same horizon.

3.7 Skill_{VP} as a diagnostic auxiliary metric

RMSE-based skill and variance retention capture different aspects of forecast behavior: the former summarizes average squared deviations, while the latter summarizes amplitude preservation. To characterize horizons where persistence-relative RMSE skill coexists with low variance retention, we define Skill_{VP} as a variance-preserving diagnostic auxiliary:

$$\text{Skill}_{VP}(h) = \text{Skill}_{\text{RMSE}}(h) \times \min\left(1, \frac{\text{Var}_{\text{pred}}(h)}{\text{Var}_{\text{obs}}(h)}\right). \quad (7)$$

The $\min(\cdot)$ bound prevents inflation when predictions are over-variable. This construction is intentionally simple and is scoped narrowly: Skill_{VP} is a horizon-wise diagnostic complement to RMSE-based skill, not a candidate replacement for it, and it makes no claim about probabilistic calibration or decision-theoretic optimality. Its sole purpose is to flag, at a glance, horizons where nominal skill and amplitude preservation diverge.

3.8 Train-calibrated P75 event diagnostics

As a complementary event-based check, an event is defined as $y_{t+h} > q_{0.75}^{(f)}$, where $q_{0.75}^{(f)}$ is the 75th percentile of the valid PM₁₀ observations in fold f 's training window. The same fold-specific threshold is applied to observations and predictions in that fold's test window. We report recall, precision, prediction flag rate, and observed event base rate separately by horizon. Precision is undefined when a model issues no positive flags; it is retained as missing rather than replaced by zero. P75 is a data-derived diagnostic threshold, not a regulatory exceedance limit.

4 Results

4.1 E1: Canonical horizon-wise results under rolling-origin evaluation

Table 2 and Figure 1 report the leakage-free rerun. LightGBM does not consistently improve on persistence: its skill is -0.010 , 0.073 , -0.034 , and -0.079 at $h = 1, 6, 24, 48$. SARIMA has modest positive skill at all four horizons (0.031 – 0.132), but its variance retention falls sharply with lead time, from 79.8% at $h = 1$ to 35.7% at $h = 6$, 2.9% at $h = 24$, and 0.4% at $h = 48$. Thus the diagnostic issue is not a comparison of two models that both outperform persistence; it is the narrower observation that SARIMA's best relative skill occurs where its predictive amplitude is least credible.

Skill_{VP} makes this decoupling explicit per horizon by attenuating persistence-relative RMSE skill when variance retention is low (Figure 2, left). For SARIMA, Skill_{VP} falls from 0.025 and 0.029 at $h = 1$ and 6 to 0.001 and effectively zero at $h = 24$ and 48 . LightGBM's negative skill is not rescued by variance retention: Skill_{VP} remains non-positive except at $h = 6$ (0.040).

Importantly, the LightGBM results do not support a strong narrative of long-horizon variance collapse within the currently evaluated range. Variance retention is non-monotone for LightGBM and reaches 86.6% at $h = 48$. The stronger diagnostic signal is model-dependent: SARIMA provides the clearer extreme case, while LightGBM illustrates that amplitude preservation alone cannot compensate for a lack of baseline-relative accuracy.

4.2 E2: Protocol sensitivity (rolling-origin vs. blocked holdout)

As a secondary sensitivity check, we report an analogous chronological 80/20 blocked-holdout evaluation. It is not treated as a substitute for the canonical rolling-origin analysis. The holdout results support the same qualitative long-horizon SARIMA pattern. At $h = 24$ and 48 , its variance retention is 3.35% and 0.07% , while skill is -0.029 and 0.029 . LightGBM holdout skill is -0.065 , 0.057 , -0.062 , and 0.066 , with variance retention of 79.5% , 62.9% , 68.2% , and 32.6% . The sign and magnitude of skill are protocol-sensitive, but SARIMA's long-horizon amplitude attenuation appears under both evaluations.

Table 2: Canonical horizon-wise results under rolling-origin evaluation (E1). VR denotes variance retention (%), and Skill_{VP} is a diagnostic auxiliary adjustment of persistence-relative RMSE skill by variance retention.

h	LightGBM				SARIMA			
	RMSE	Skill	VR (%)	Skill _{VP}	RMSE	Skill	VR (%)	Skill _{VP}
1	5.667	-0.010	68.9	-0.007	5.433	0.031	79.8	0.025
6	9.147	0.073	55.1	0.040	9.050	0.083	35.7	0.029
24	12.802	-0.034	59.9	-0.021	11.819	0.045	2.9	0.001
48	15.228	-0.079	86.6	-0.069	12.241	0.132	0.4	0.000

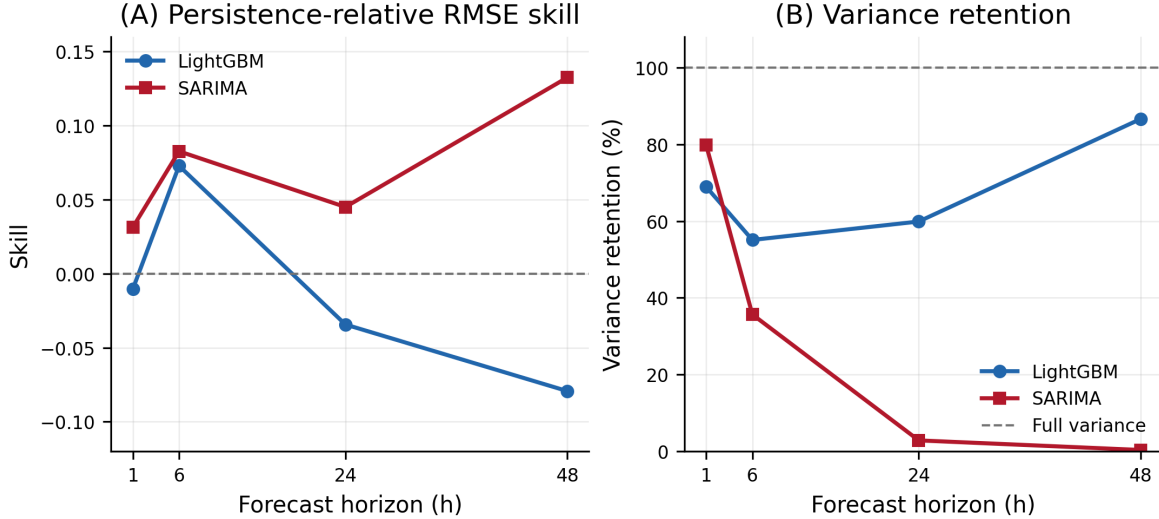


Figure 1: Rolling-origin horizon-wise results for LightGBM and SARIMA. *Left*: persistence-relative RMSE skill. *Right*: variance retention (%). SARIMA retains modest positive skill while its variance collapses at long horizons; LightGBM does not consistently beat persistence.

4.3 E3: Event-oriented complement

As a supportive event-oriented complement, we evaluate detection at the P75 threshold using one threshold fitted on each rolling-origin fold’s training data. LightGBM recall decreases from 0.773 at $h = 1$ to 0.323 at $h = 48$, with precision decreasing from 0.815 to 0.390. SARIMA recall decreases from 0.785 at $h = 1$ to 0.099 at $h = 24$ and zero at $h = 48$; its flag rate likewise falls from 0.300 to zero. Its precision at $h = 48$ is undefined, rather than zero, because the model issues no positive flags. This long-horizon loss of event sensitivity is consistent with its near-zero variance retention.

These event-oriented results are treated as supportive diagnostics. Recall alone can be trivially high under threshold saturation—when forecasts remain above a permissive threshold at most time steps—and should therefore be interpreted jointly with precision, flag rate, and variance retention. Under fold-recalibrated thresholds, neither model sustains strong event detection across all horizons, and SARIMA does not flag P75 events at $h = 48$.

Overall, E1–E3 jointly support a bounded, model-dependent conclusion: persistence-relative RMSE skill can remain positive while dynamic fidelity differs markedly across models, and variance retention together with Skill_{VP} provides a practical diagnostic view of this divergence for horizon-wise interpretation.

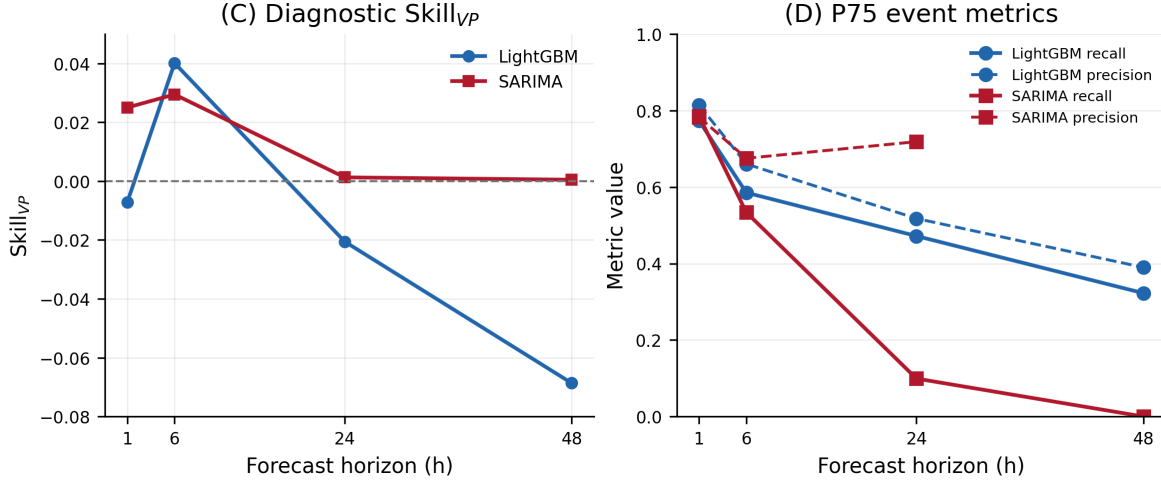


Figure 2: Diagnostic complements for horizon-wise interpretation. *Left*: Skill_{VP}, which adjusts persistence-relative RMSE skill by variance retention. *Right*: horizon-wise P75-event recall and precision under fold-recalibrated evaluation. SARIMA precision at 48 h is undefined and is omitted because the model issues no positive flags.

5 Discussion

The evidence in this manuscript supports a diagnostic, horizon-wise interpretation of multi-horizon PM₁₀ point forecasts under rolling-origin evaluation with persistence as an explicit baseline. LightGBM does not consistently improve on persistence, whereas SARIMA remains modestly positively skilled in the rolling-origin evaluation. The variance diagnostics show that SARIMA’s positive skill at $h = 24$ and 48 coincides with near-collapsed predictive variance. This result is consistent with the view that even a positive error-based skill value can mask severe loss of amplitude, and that horizon-wise interpretation changes when variance retention is examined alongside RMSE-relative skill.

Skill_{VP} provides a compact auxiliary description of this divergence across horizons; it does not establish predictive usefulness or replace the underlying RMSE, variance, and event metrics. The LightGBM results do not support a monotone variance-collapse narrative, although its negative skill at three horizons rules out claims of consistent improvement over persistence. SARIMA provides clearer evidence that modest RMSE-relative skill can coexist with near-zero variance retention, and the event-oriented evaluation supplies a convergent diagnostic at long leads.

Event-oriented evaluation must be handled cautiously. Recall alone may be trivially high under threshold saturation and does not by itself demonstrate predictive discrimination or operational usefulness. In this study, fold-specific training thresholds avoid using future test values to define events, and recall is reported jointly with precision and flag rate. Under these conditions, the event metrics support the same cautious conclusion as the variance diagnostics: performance deteriorates with horizon for both models, and SARIMA’s positive long-horizon skill coexists with no positive P75 flags at 48 h.

6 Limitations

This study is intentionally limited in scope. The empirical evidence is anchored in a single station, a single calendar year, and five rolling-origin folds; external validity to other pollutants, temporal resolutions, monitoring sites, or longer evaluation windows has not been established. The results are restricted to the evaluated horizons $h \in \{1, 6, 24, 48\}$ and to two model classes (LightGBM, SARIMA) with specific, fixed configurations; neither model was tuned to compete for state-of-the-art accuracy, since the goal is to characterize a decoupling pattern rather than to benchmark model performance. The study reports point estimates without confidence intervals or formal tests of forecast-loss differences. The P75 analysis measures

sensitivity to a training-calibrated distributional threshold and must not be interpreted as performance against a legal air-quality threshold. Skill_{VP} has not been validated outside the present case study, and its behavior under regimes with strong seasonality, structural breaks, variance inflation, or decision-specific losses remains untested.

7 Conclusion

In this PM_{10} case study evaluated under rolling-origin protocols with persistence as an explicit baseline, the reproducible evidence is modest: LightGBM does not consistently beat persistence, and SARIMA achieves only small positive skill. Nevertheless, SARIMA’s variance retention falls to 2.9% and 0.4% at 24 and 48 hours, and its P75-event recall falls to 0.099 and zero. Reporting variance retention alongside RMSE skill, and using Skill_{VP} as a diagnostic auxiliary metric, offers a practical way to make this divergence explicit for horizon-wise interpretation under rolling-origin evaluation, without implying a universal replacement for established verification practices.

Data and Code Availability

The processed series, row-level rolling-origin and holdout predictions, aggregate metrics, run manifests, analysis code, and evaluation scripts used in this study are available at <https://github.com/fedeg-umh-es/varret-pm10-paper>. Raw PM_{10} data were obtained from the Madrid Open Data hourly air-quality archive (<https://datos.madrid.es/dataset/201200-0-calidad-aire-horario>) and are available under CC BY 4.0. The repository records the annual archive URL and SHA-256 checksum and includes scripts that recover the station series and reproduce predictions, metrics, event diagnostics, and figures.

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